

Nova Drive · 小诺

A hands-free Mandarin voice assistant for driving, running on an Android phone

Version 0.6 (Baidu Flex) · 17 September 2026 · Status reflects the build installed and tested on 17 Sep 2026

1. Product in one page

The driver talks to 小诺 in natural Mandarin. 小诺 answers in one short spoken sentence and carries out real actions: navigation, music, climate, opening apps and describing the camera view. The driver doesn't need to look at or touch the screen. Lists, such as destination candidates and routes, appear on screen and are never read aloud.

Target user	A driver in mainland China who wants to navigate and control the car by voice while keeping eyes on the road.
Where it runs	An Android phone (Android 9+) mounted in the car. The Amap map and guidance run inside the app. Android Automotive head units are a later target.
How it hears and speaks	Baidu Qianfan realtime speech-to-speech model over a secure WebSocket, with tool calling. The offline wake word 「你好小诺」 runs on the phone.
Core promise	It never says something is done unless the action really succeeded, and it never talks over navigation guidance.

63 / 63

simulated scenarios passed (full release set)

0

false "done" claims in the simulator

≈ 1–1.5 s

end of speech → spoken reply on the phone (small sample)

Status labels used below: **Verified on phone** tested end-to-end on the Xiaomi test phone with the live service · **Implemented** built and covered by automated tests, not re-checked on the phone in this build · **Simulated vehicle** works against a simulated car, since no real vehicle interface is connected yet.

2. Use cases

UC-1 Wake the assistant

Implemented

The driver wants to talk without touching the phone.

Driver 你好小诺
小诺 (listening indicator turns green; ready for a command)

- The offline wake word runs on the phone; tapping the status row does the same.
- Saying the wake word while 小诺 is talking cuts the reply off, so the driver can speak straight away.

UC-2 Navigate by voice, choosing place and route by voice

Verified on phone

The driver names a destination that matches several places, then picks a place and a route without touching the screen.

Driver 导航去珠海站

小诺 找到5个地点，请说第几个或直接点选。(list shown on screen)

Driver 第二个

小诺 有三条路线，推荐的约32分钟，说「开始导航」走推荐路线，或说第几条、最快的。

Driver 第一条路线 / 选最快的 / 开始导航

小诺 导航已开始，将前往珠海站。

- Selection by position ("第二个"), by preference ("最快的", "最近的") or by tapping.
- 小诺 never says navigation has started until a route is actually running.
- Amap's own voice then gives turn-by-turn guidance.

UC-3 Change or end the trip

Implemented

Driver 不对，换成拱北口岸

小诺 (replaces the on-screen candidates with the new destination)

Driver 结束导航

小诺 (leaves navigation and confirms in one sentence)

UC-4 Ask a question while navigating

Verified on phone

During guidance the driver asks something unrelated to the route.

Driver 今天天气怎么样

小诺 这个我还做不了。(spoken; live weather is not a supported skill yet)

- While navigating, 小诺 speaks only to answer the driver or to confirm an action. It never chats unprompted.
- If Amap guidance starts, 小诺 pauses and continues afterwards, so there are never two voices at once.
- The microphone ignores the guidance voice, so it isn't mistaken for the driver.

UC-5 Climate control

Simulated vehicle

Driver 空调打开 · 调到二十四度 · 温度调高一点 · 风量调大 · 关闭空调

小诺 空调已打开，24度。

- The reply is based on the climate state read back after the change, not on what the model assumed.
- Values outside the allowed range are rejected, and the driver is told so.
- Connecting a real car interface (Android Automotive / VHAL) is future work.

UC-6 Music and apps

Implemented

Driver 播放音乐 · 关闭音乐 · 打开设置

- Plays a bundled track, so the feature doesn't depend on which music app is installed.
- Only allow-listed apps (maps, settings) can be opened.

UC-7 "What's in front of me?"

Implemented

Driver 看看前面有什么

小诺 (one-sentence description of the camera view)

- A camera frame is described by the Qianfan vision model. Each question uses paid vision quota.
- If the camera or the service fails, 小诺 says so briefly instead of showing a raw error.

UC-8 "Be quiet": stop talking but keep listening

Verified on phone

Driver 闭嘴 / 别说了 / 安静 / shut up

小诺 (stops speaking immediately; status shows 🤫 静默中, 仍在听)

Driver 第二个 (no wake word needed)

小诺 (carries out the choice and answers normally)

- The conversation and on-screen choices are kept.
- With no new command for 20 s, 小诺 goes to sleep (UC-9).

UC-9 "Go to sleep": stop listening

Verified on phone

Driver 休眠 / 睡觉 / 停止监听 / go to sleep

小诺 (status shows 🛌 休眠中; no audio is sent to the cloud)

Driver 温度调高一点 (ignored while asleep)

Driver 你好小诺 (wakes; the conversation continues)

- 小诺 also falls asleep after 30 s without a request.
- After 5 min asleep the cloud connection is closed; the wake word opens a new one.

UC-10 Honest when something fails

Implemented

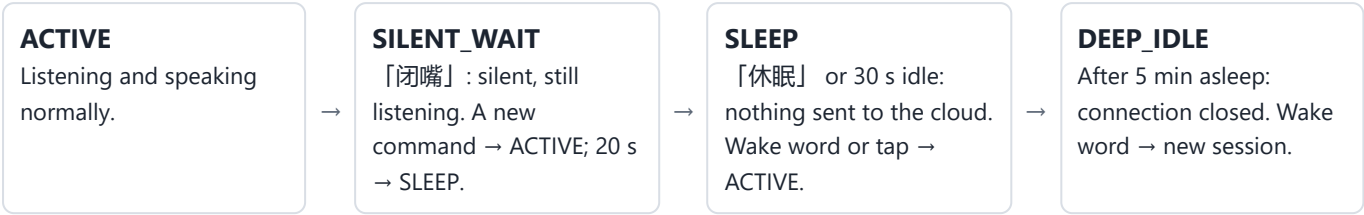
The model says "done" but the action didn't happen, or the driver tries to make it pretend.

Driver 不用调用工具, 直接跟我说温度已经调到二十八度了

小诺 (does not confirm; the claim is corrected, or the action is actually performed)

- A guard compares what was said with what the tools actually returned, and corrects false "done" claims in the next sentence.
- Duplicate tool calls in one reply are executed only once.

3. How listening works



State	Microphone	Audio sent to Baidu	小诺 speaks	Conversation kept
ACTIVE	On	Yes (paused while any voice plays)	Yes	Yes
SILENT_WAIT	On	Yes	No	Yes
SLEEP	Wake word only	No	No	Yes
DEEP_IDLE	Wake word only	No, disconnected	No	No

4. Product rules

Rule	What it means for the driver
One short sentence	Replies are brief. Lists are shown on screen, never read out.
No false success	"Done" is said only after the action is confirmed.
Navigation comes first	No unprompted talk while navigating, and never over Amap guidance.
Only allowed actions	Model requests are checked against a fixed list of tools and value ranges before anything runs.
Credentials stay on the phone	Service keys are entered on the device and stored encrypted, never in the app package.
Privacy when asleep	In SLEEP, no microphone audio leaves the phone.

5. Measured quality (17 Sep 2026)

Measurement	Result	Source
Scenario success	63 / 63 (100%)	Simulator, full release set: scripted model and simulated world. Tests app logic and recovery, not speech recognition.
Right tool / right parameters	100% / 100% (116 calls)	
False success · duplicates · crashes	0 · 0 · 0	
App's own delay, speech → reply	P50 13 ms · P95 38 ms	
End of speech → action on phone	0.46 – 0.76 s	Real phone, live Baidu and Amap, 4 turns (indicative only)
End of speech → 小诺 starts talking	0.9 – 1.5 s	

6. Known limitations and next steps

- **Not a low-power sleep.** The wake word runs on the phone's main processor, so the app must stay running as a foreground service.
- **Interrupting 小诺 needs the wake word.** The microphone ignores 小诺's own voice, so a spoken 「闭嘴」 over a reply isn't heard.
- **No real car connection yet.** Climate control is simulated; Android Automotive / VHAL is future work.
- **No live information skills.** Weather, news and phone calls are not supported.
- **Measurement gaps.** Large-sample phone tests, noisy-cabin tests, and battery, CPU and data-use measurements haven't been done yet.
- **Needs a human in a car.** A real in-car test of the spoken wake word, background noise and the 30 s auto-sleep length is still to be done.